VL **Venkat Loganathan**
PROMPT ENGINEERING · INSTRUCTOR

COURSE 23DS1D0
ONE-CREDIT · 15 HOURS

Prompt Engineering — What to Expect

A hands-on, 2-day workshop for M.Sc (Data Science & AI) students

Over two days you'll learn to get **reliable, useful work out of AI models** — not by luck, but by method. This is a workshop, not a lecture series: for most of our time your laptop is open and you're building, testing, and improving prompts. You'll leave able to design prompts that work, and prove that they work.

DAY 1 Saturday, 18 July 2026	DAY 2 To be confirmed	TIMING 9:00 – 17:00 (45-min lunch)
VENUE College campus (room TBA)	FORMAT ~60% hands-on labs	COST TO YOU Free tools only

What you'll learn

CO1 **Design & optimize prompts.** Write clear, structured prompts that behave consistently across tasks and models.

CO2 **Assess & evaluate.** Test prompts for performance, ground them in real data, and check them for security and safety.

Concretely, you'll be able to: apply few-shot and chain-of-thought prompting; force reliable structured (JSON) output; break hard problems down and use self-consistency; build a small tool-using agent; ground answers in documents with retrieval (RAG); and red-team and benchmark your own prompts.

What's covered

1	**Introduction to Prompts**	Terminology, how LLMs work — tokens, temperature, context window, why they hallucinate.
2	**Text-Based Techniques**	In-context learning, zero-shot, few-shot, chain-of-thought reasoning.
4	**Answer Engineering**	Shaping output, constraining answers, reliable JSON & extraction.
3	**Advanced Techniques**	Decomposition, ensembling / self-consistency, self-criticism.
5	**Multilingual & Extensions**	Tool-use agents, code-generation agents, multilingual prompting.
6	**RAG, Security & Benchmarking**	Grounding & citations, prompt injection & jailbreaks, evaluation.

How the labs run

- **Learn by doing.** Every concept is followed immediately by a short lab. Nine labs in total, building up to a small team capstone on Day 2.
- **Three platforms.** You'll use ChatGPT and Claude (free web versions) and a local open-source model via **Ollama** — so you compare results and nothing costs money.
- **Pair-friendly.** You may work in pairs, especially during setup. Sample data files are provided for every lab.
- **Assessment:** Labs 40% · take-home assignment 35% · capstone + short viva 25%. No written exam.

What you need to arrange — before Day 1

PRE-CLASS CHECKLIST (PLEASE COMPLETE BEFOREHAND)

- **Bring a laptop** (Windows / Mac / Linux) with at least **8 GB RAM** and ~5 GB free disk space, plus its **charger**.
- **Install Ollama** from **ollama.com**, then run `ollama pull llama3.2` and `ollama pull mistral` (these downloads are large — do them on home/hostel WiFi in advance).
- **Create free accounts** at **chat.openai.com** (ChatGPT) and **claude.ai** (Claude).
- **Have Python 3** installed (basic familiarity is enough; starter code is provided).
- **Come curious.** Bring a real task you'd like an AI to help with — we may use it in a lab.

Questions before the session? Reach me directly.

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